

W241 Plank Study Experimental Report

Crossover and benchmark analysis

W241 Plank Study

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0.1 Report Structure

This report follows the order in the experiment analysis plan:

1. participant flow, attrition, and exclusions
2. primary crossover effect
3. period and order diagnostics
4. first-period benchmark
5. realized precision comparison
6. secondary outcomes and HTE
7. randomization inference

0.2 Study Overview

The primary estimand is the **within-person effect of high-tempo versus low-tempo music on plank time-to-exhaustion**, measured in seconds. The main estimate uses the randomized crossover design, so each participant serves as their own control and stable participant-level differences are absorbed by construction.

The first-period benchmark is supplementary. It asks what the treatment contrast would have looked like if the study had been analyzed as a standard parallel-arm design using only session 1. That benchmark is included to clarify the design choice, not to replace the crossover estimate.

0.3 Participant Flow And Analysis Populations

The report begins with the full registered participant roster, then narrows to observed outcomes and protocol-valid analytic samples. This keeps attrition, missingness, and protocol exclusions visible before any inferential models are introduced.

0.4 Reading the flow correctly

This step intentionally starts from the **full initial distribution** and only then narrows to the analytic samples.

- First, we show the full registered roster and how many participants reach each observation stage.
- Next, we separate **attrition** (for example, not returning for session 2) from **compliance / protocol exclusions**.
- Only after those counts are visible do we define the **primary analytic sample** and the **first-period benchmark sample**.

This mirrors the report plan: the reader should see the whole study population first, then exactly why some participants are not used for a given estimand.

0.5 Build participant flow dataset

The participant roster is the base denominator for flow accounting. We left-join the experiment wide file onto the full participant list so registered participants with zero observed sessions still appear in the flow dataset.

0.6 Participant flow summaries

Table 1: Table A. Participant flow

population	n
Registered participants	58
Randomized order assigned	58
Session 1 observed	47
Session 2 observed	40
Completed both sessions	40
Primary analytic sample	38
First-period benchmark sample	46

Participant flow from registration to analytic samples

Full distribution first, then attrition and protocol-based exclusions

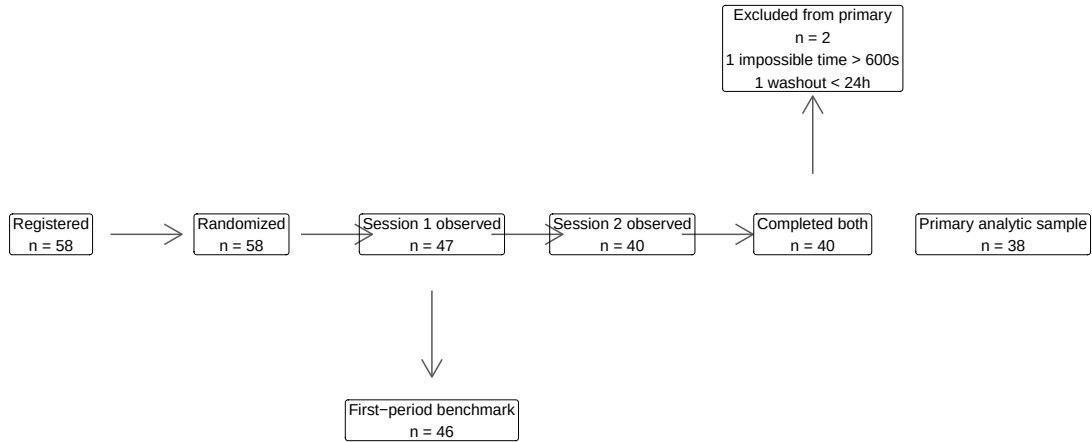


Table 2: Table B. Attrition by randomized order

assigned_order	started_session1	dropped_before_session2	retained_completed_both
high_tempo_first	23	4	19
low_tempo_first	24	3	21

0.7 Attrition-bias check

This check asks whether retention through both observed sessions differed meaningfully by randomized order arm or baseline demographics among randomized participants. It is intentionally light-touch: the goal is to surface obvious selection patterns without overfitting a small attrition model.

Table 3: Table C. Baseline demographics by retention status

retained_group	n	mean_age	sd_age	share_male
Completed both sessions	40	36.30	13.15	70.0
Did not complete both	18	32.67	10.05	66.7

Table 4: Table D. Retention by randomized order

assigned_order	randomized_participants	retained_completed_both	not_retained	retention_rate_pct
high_tempo_first	30	19	11	63.3
low_tempo_first	28	21	7	75.0

Table 5: Table E. Attrition-bias diagnostic p-values

comparison	p_value
Retention by randomized order	0.4022
Baseline age by retention status	0.2556
Baseline gender by retention status	1.0000

These diagnostics should be read as descriptive checks rather than proof that attrition was ignorable. They focus on retention through Session 2 among randomized participants and cannot rule out selection on unobserved factors. They are also conceptually separate from the protocol-based primary-analysis exclusions, which apply only after a participant completed both observed sessions.

Table 6: Table F. Compliance and exclusion reasons

issue	n
Impossible time > 600s	1
Washout violation < 24h	1
Missing one condition	18
Excluded from primary analysis	2

0.8 Build paired analytic dataset

The paired dataset keeps one row per primary analytic participant and standardizes condition-level secondary outcomes so later Rmds can work in `high_tempo` / `low_tempo` terms directly.

0.9 Build long repeated-measures dataset

The long dataset keeps one row per observed session and carries analysis flags so later model files can subset consistently without re-implementing the exclusion logic.

0.10 Write derived outputs

0.11 Primary Crossover Estimate

With the populations fixed, the next step is the primary within-subject treatment estimate. The emphasis here is the design-consistent sequence: paired descriptives first, then the paired mean difference, then a participant fixed-effects model without period adjustment, and finally the fixed-effects crossover model with a period adjustment.

0.12 Analytic sample

Table 7: Primary crossover sample

quantity	value
Paired analytic participants	38
Repeated-measures rows used in FE model	76
High-tempo first	19
Low-tempo first	19

0.13 Descriptive summaries

Table 8: Paired descriptive summaries

quantity	value
Mean high-tempo TTE	102.440
Mean low-tempo TTE	88.071
Mean paired difference (high - low)	14.369
SD paired difference	28.921
Median paired difference	8.708
IQR paired difference	25.199
Independent participants	38.000

0.14 Compact metrics summary

Table 9: Requested summary metrics

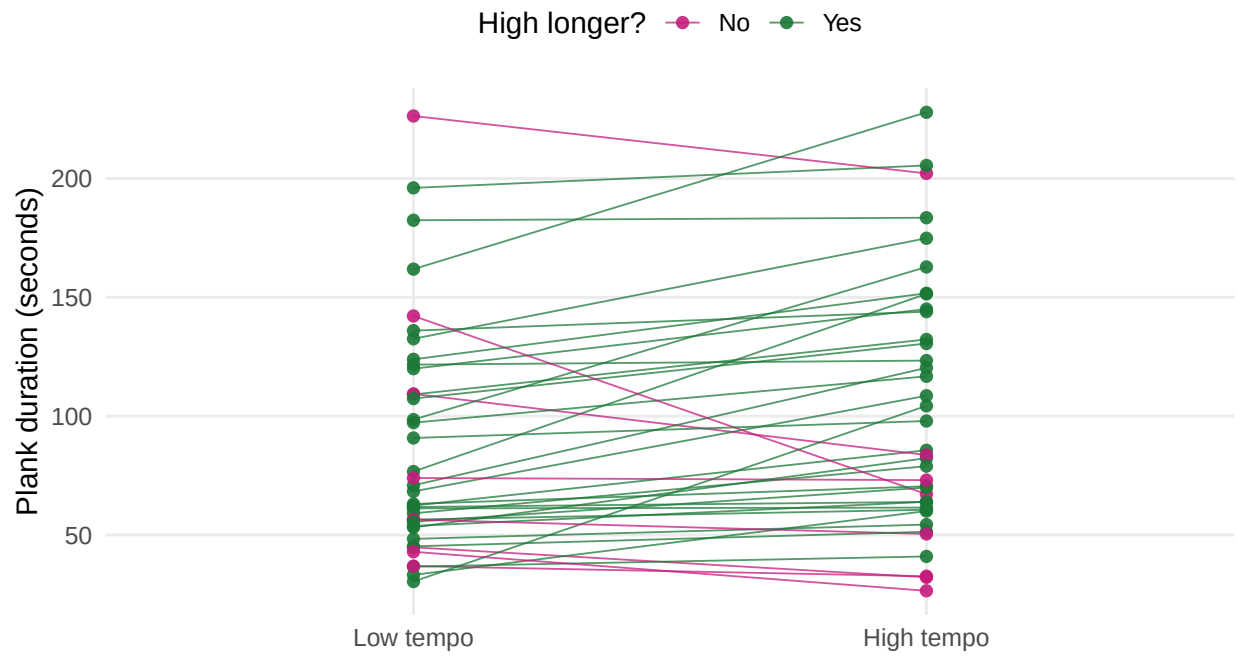
metric	value
Mean treatment effect (high - low, seconds)	14.37
Percentage improvement vs low tempo mean	16.3%
Standard deviation of the paired difference	28.92
Paired t-test p-value	0.0041
Completion rate (completed both / registered)	69.0%

0.15 Exploratory figures

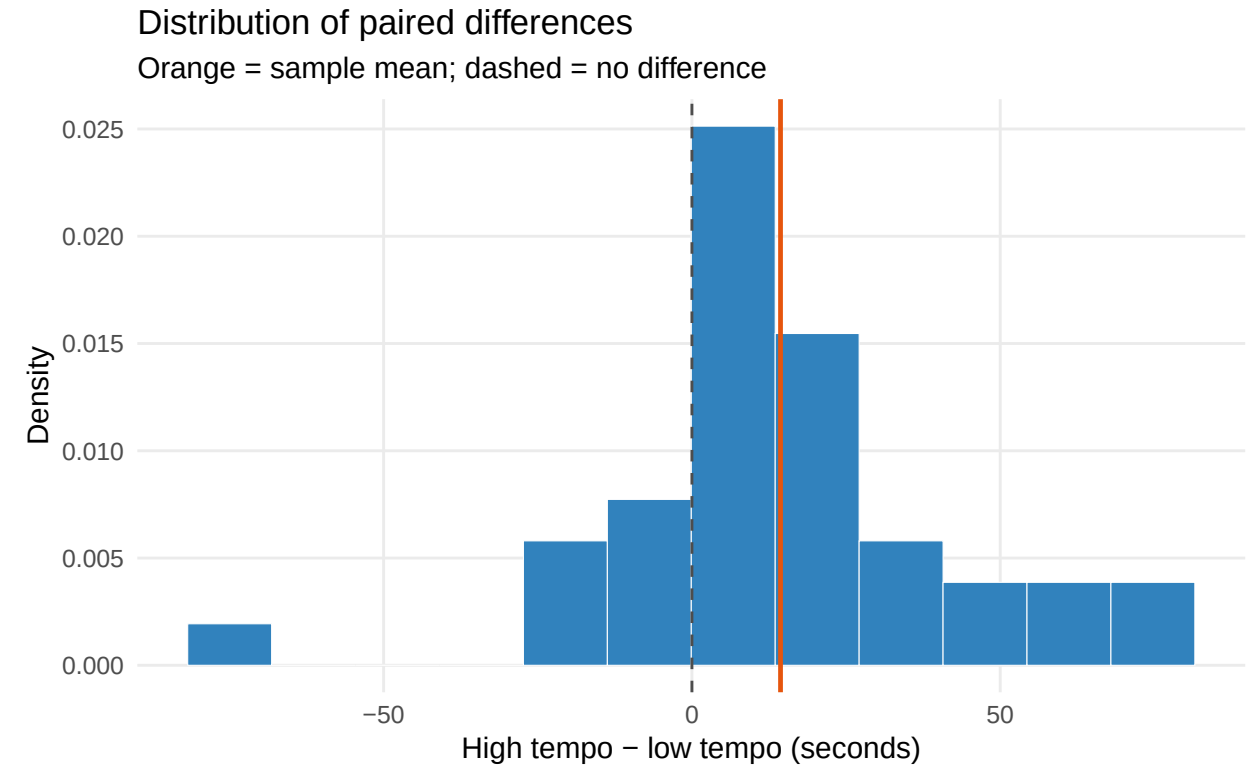
0.15.1 Spaghetti plot

Paired plank times by music condition

n = 38 primary-sample participants



0.15.2 Paired-difference distribution



0.16 Model 1: raw paired contrast

Table 10: Model 1. Raw paired contrast

quantity	value
Mean high-tempo TTE	102.4404
Mean low-tempo TTE	88.0709
Mean effect (high - low)	14.3694
95% CI lower	4.8633
95% CI upper	23.8756
Paired t statistic	3.0628
Paired p-value	0.0041
Standard error	4.6916
Paired effect size dz	0.4968

0.17 Model 2: fixed-effects crossover regression without period adjustment

Table 11: Model 2. Fixed-effects crossover regression without period adjustment

term	estimate	std_error	conf_low	conf_high	p_value
High-tempo indicator	14.3694	6.6796	1.2774	27.4615	0.0381

0.18 Model 3: fixed-effects crossover regression with period adjustment

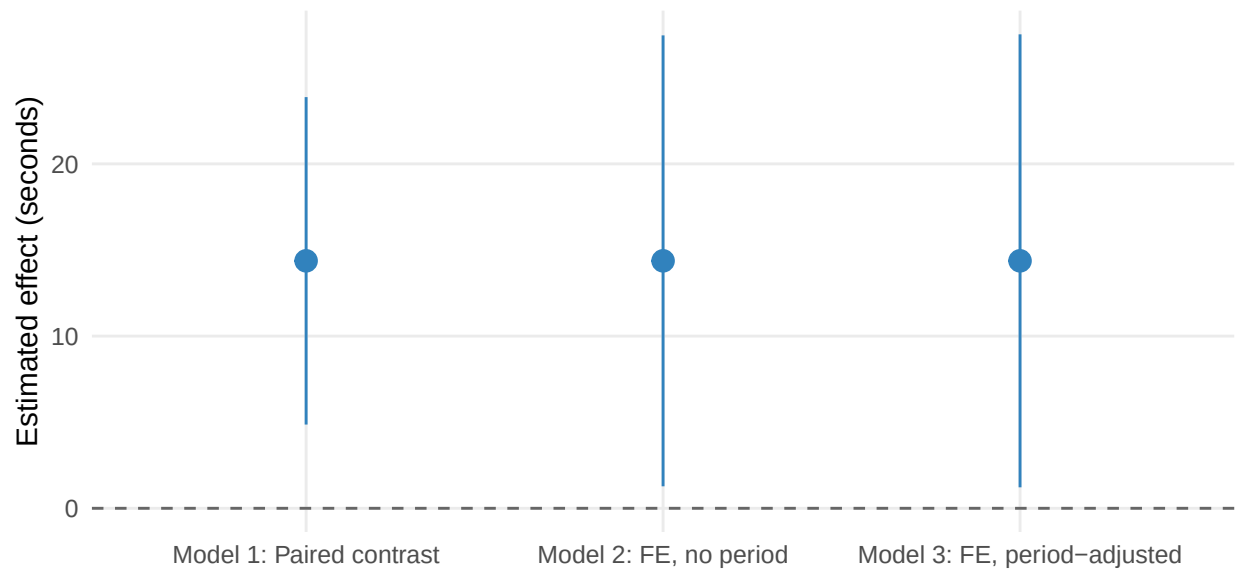
Table 12: Model 3. Fixed-effects crossover regression with clustered SEs

term	estimate	std_error	conf_low	conf_high	p_value
High-tempo indicator	14.3694	6.7101	1.2177	27.5212	0.0391
Period 2 indicator	3.8441	6.7101	-9.3076	16.9958	0.5703

0.19 Main-effect comparison figure

Primary effect estimates

Effect on plank time in seconds (high tempo – low tempo) across the three-model ladder



0.20 Period And Order Diagnostics

After the main estimate, the report checks whether broad period effects or order patterns appear large enough to complicate interpretation. These diagnostics are supportive rather than alternative estimators.

0.21 Period summaries

Table 13: Session 1 vs Session 2 summaries

period	n	mean_tte	sd_tte	median_tte	iqr_tte
1	38	93.334	46.730	84.646	68.303
2	38	97.178	55.207	70.692	75.268

0.22 Period-by-order summaries

Table 14: Period-by-order diagnostic summary

crossover_order	period	treatment	n	mean_tte	sd_tte	mean_rpe
high_tempo_first	1	high_tempo	19	91.001	44.336	6.895

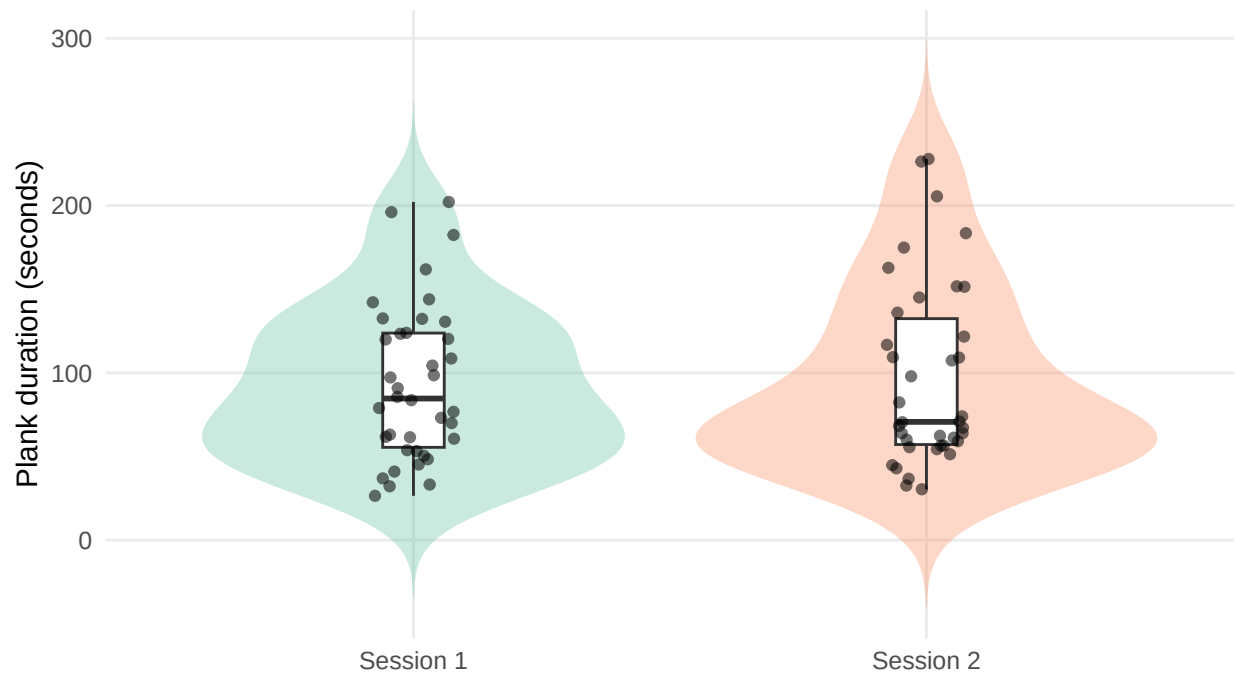
crossover_order	period	treatment	n	mean_tte	sd_tte	mean_rpe
high_tempo_first	2	low_tempo	19	80.476	46.395	6.263
low_tempo_first	1	low_tempo	19	95.666	50.116	7.105
low_tempo_first	2	high_tempo	19	113.880	59.359	7.632

0.23 Session-period figures

0.23.1 Session 1 vs Session 2 boxplot

Outcome distributions by session period

Primary analytic sample, n = 38



0.23.2 Order-arm trajectories across periods

Average plank time by order arm across periods

Checks for sequence or learning patterns



0.24 Diagnostic models

0.24.1 Broad period check

Table 15: Broad period comparison

quantity	value
Mean Session 2	97.1777
Mean Session 1	93.3336
Difference (Session 2 - Session 1)	3.8441
95% CI lower	-19.5456
95% CI upper	27.2339
p-value	0.7441

0.24.2 Sequence diagnostic regression

Table 16: Sequence diagnostic regression

term	estimate	std_error	p_value
period_factor2	-10.5253	5.8264	0.0750
crossover_orderlow_tempo_first	4.6650	15.4541	0.7636
period_factor2:crossover_orderlow_tempo_first	28.7389	9.4895	0.0034

0.25 Interpretation notes

0.26 First-Period Benchmark

Only after the primary crossover evidence and the period/order checks do we turn to the first-period benchmark. This keeps the benchmark in its intended role: a secondary comparison that illustrates what a parallel-arm analysis would have looked like before carryover could matter.

0.27 Benchmark sample summary

Table 17: First-period benchmark sample

quantity	value
Benchmark participants	46
Assigned high tempo first	23
Assigned low tempo first	23

0.28 Balance checks

Table 18: Compact randomization check artifact: counts, age, gender, and baseline activity

treatment_label	n	mean_age	share_male	share_active_at_least_weekly
High tempo first	23	31.17	73.9	34.8
Low tempo first	23	40.35	60.9	47.8

Table 19: Balance check: age by assigned first condition

treatment_label	n	mean_age
High tempo first	23	31.174
Low tempo first	23	40.348

Table 20: Balance check: gender counts

treatment_label	female	male
High tempo first	6	17
Low tempo first	9	14

Table 21: Balance check: baseline activity counts

treatment_label	1	2	3	4
High tempo first	9	6	1	7
Low tempo first	7	5	0	11

The first-period benchmark arms are balanced in **counts** but not perfectly identical on observables. The main age difference is visible here, which is worth acknowledging. It is less threatening for the primary estimate because the main analysis uses the crossover design and participant fixed effects, which absorb stable participant-level differences.

0.29 Benchmark descriptives

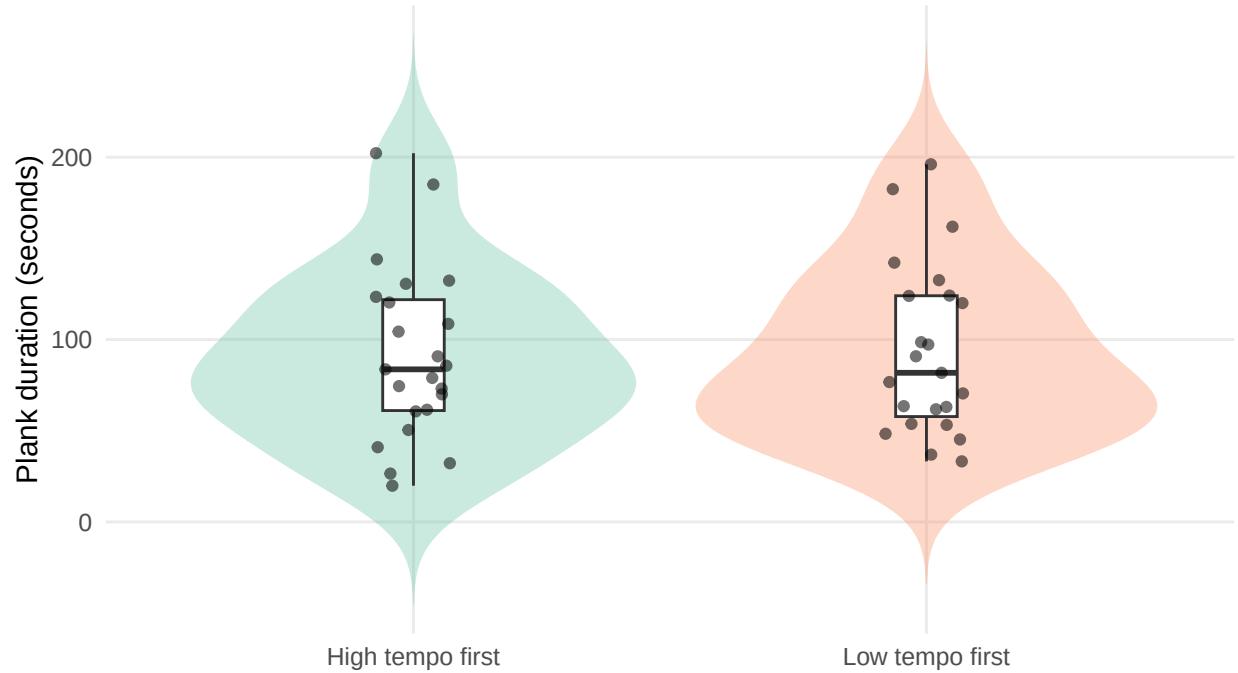
Table 22: Session 1 descriptives by assigned condition

treatment_label	n	mean_tte	sd_tte	median_tte	iqr_tte
High tempo first	23	91.265	47.460	83.655	60.795
Low tempo first	23	93.801	46.614	81.760	66.236

0.30 Figure

Session 1 benchmark by assigned first condition

n = 46 independent participants



0.31 Benchmark Model 1: raw contrast

Table 23: Benchmark Model 1. Raw period-1 contrast

quantity	value
Mean high-first session 1 TTE	91.2654
Mean low-first session 1 TTE	93.8013
Difference (high first - low first)	-2.5359
95% CI lower	-25.4194
95% CI upper	30.4912
t statistic	0.1828
p-value	0.8558

0.32 Benchmark Model 2: precision-adjusted regression

Table 24: Benchmark Model 2. Precision-adjusted regression

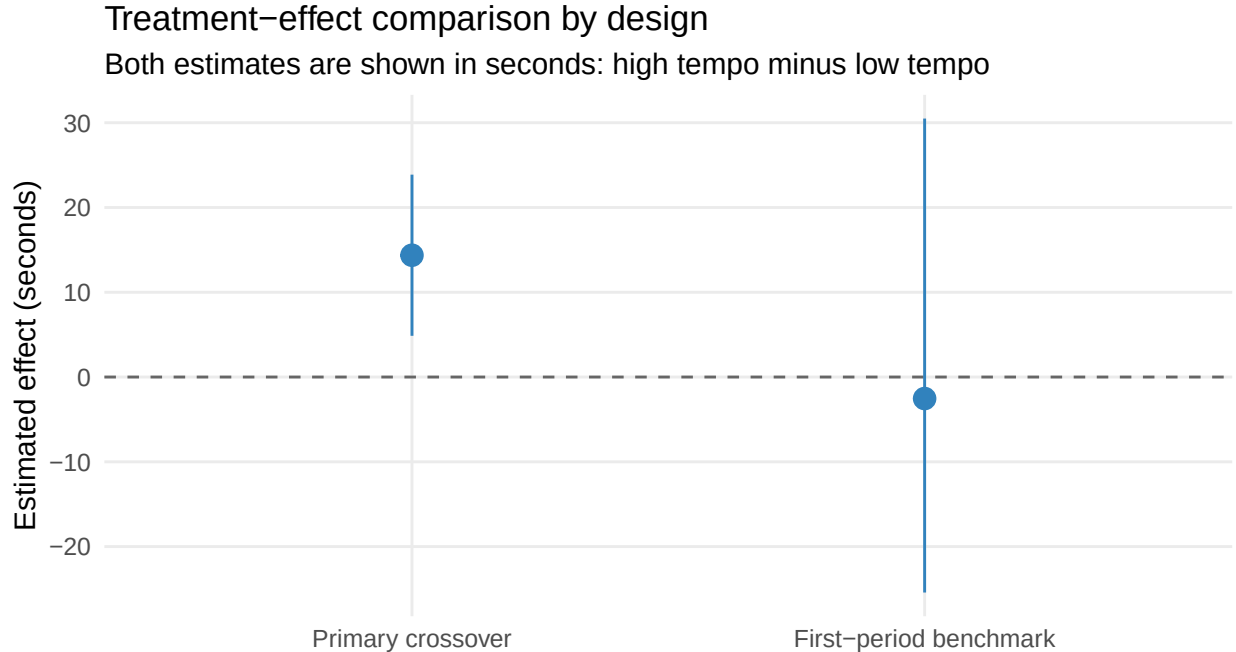
term	estimate	std_error	p_value
High-tempo first	-6.6796	15.0537	0.6595
Participant age	-0.1142	0.6001	0.8499

0.33 Results At A Glance

The key treatment-effect results are consolidated here so the reader can compare the primary crossover estimate, the supplementary benchmark, and the design-based inference in one place.

Table 25: Compact main-results summary

statistic	value
Paired analytic participants	38
Mean low-tempo TTE	88.07
Mean high-tempo TTE	102.44
Paired effect (high - low)	14.37
Paired 95% CI	[4.86, 23.88]
Paired p-value	0.0041
FE without-period estimate	14.37
FE without-period p-value	0.0381
FE with-period estimate	14.37
FE with-period p-value	0.0391
RI two-sided p-value	0.0031



0.34 Precision And Design-Efficiency Comparison

Once both the crossover and benchmark estimates are on the page, the precision comparison shows why the crossover design is more efficient for this question.

0.35 Precision calculations

0.36 Precision table

Table 26: Realized precision by design

design	independent_n	observations	relevant_sd	standard_error	ci_width_95	mde_80_power
Primary crossover	38	76	28.9211	4.6916	18.3908	13.1440
First-period benchmark	46	46	47.0386	13.8709	54.3730	38.8606

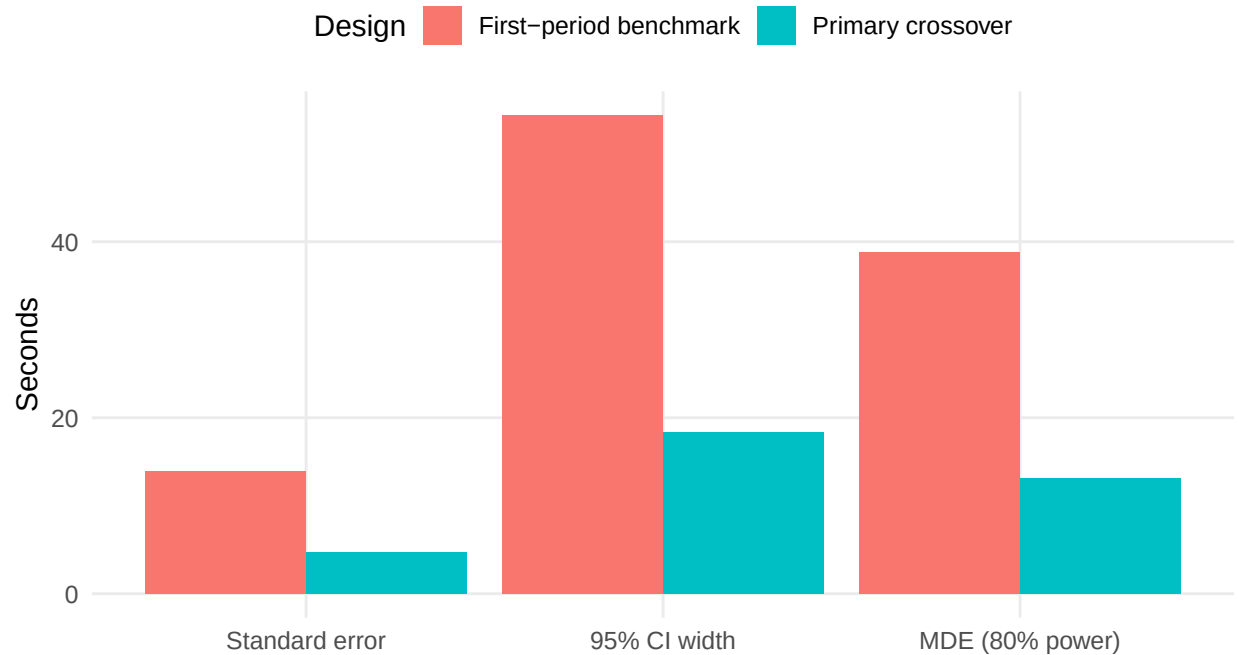
Table 27: Parallel-design sample size needed to match crossover precision

quantity	value
Observed benchmark total N	46
Required benchmark total N to match crossover SE	404
Required benchmark N per arm to match crossover SE	202
Extra independent participants needed	358

0.37 Precision comparison figure

Realized precision comparison

Smaller values indicate tighter design precision



0.38 Interpretation notes

Table 28: Relative precision loss in the benchmark design

quantity	value
Parallel SE / crossover SE	2.9565
Parallel CI width / crossover CI width	2.9565

0.39 Exploratory Mechanisms And Heterogeneity

The next section keeps the mechanism-related outcomes and subgroup patterns clearly downstream of the primary estimate. These are exploratory checks rather than headline findings.

0.40 Secondary outcomes: paired contrasts

Table 29: Exploratory paired contrasts for secondary outcomes

outcome	mean_high	mean_low	mean_diff_high_minus_low	conf_low	conf_high	p_value
RPE	7.2632	6.6842	0.5789	-0.0161	1.1740	0.0562
Motivation	5.1842	4.7105	0.4737	0.0298	0.9176	0.0371
Energy	4.6053	4.3158	0.2895	-0.1748	0.7537	0.2144

0.41 Secondary outcomes: fixed-effects versions

Table 30: Exploratory fixed-effects treatment coefficients for secondary outcomes

outcome	estimate	std_error	p_value
RPE	0.5789	0.4237	0.1803
Motivation	0.4737	0.2905	0.1117
Energy	0.2895	0.3259	0.3803

0.42 HTE setup

We estimate treatment interactions on the primary TTE outcome. The coefficient of interest in each model is the **treatment-by-subgroup interaction**, which asks whether the high-tempo effect changes across participants with different characteristics.

Because subgroup results are exploratory, we report Bonferroni-adjusted p-values across the four planned subgroup tests.

The plank-familiarity proxy comes from the question “**Do you regularly perform planks?**”. To avoid treating two slightly different “occasionally” codes as distinct levels, we collapse the raw 4-level coding into a cleaner 3-step ordinal variable:

- 1 = rarely or never
- 2 = occasionally
- 3 = regularly

0.43 HTE models

Table 31: Exploratory HTE interaction results with Bonferroni correction

Subgroup	Interaction	Estimate	SE	p	Bonferroni p
Age	High tempo x age	0.2723	0.5090	0.5960	1
Gender	High tempo x gender	5.5517	13.5212	0.6839	1
Activity level	High tempo x activity	-0.8812	5.3176	0.8693	1
Plank familiarity	High tempo x familiarity	2.8831	9.7486	0.7692	1

0.44 Design-Based Inference Check

The report closes the analytic sequence with randomization inference, which checks whether the observed paired effect is unusual under the sharp null implied by the randomization mechanism.

0.45 Observed statistic

Table 32: Observed statistic for randomization inference

quantity	value
Participants	38.0000
Observed paired mean difference (high - low)	14.3694

0.46 Bridge linear model

The randomization-inference statistic is the same paired mean difference reported in the primary analysis. To make that link explicit, we first write the observed effect as the intercept from a one-sample linear model on the participant-level differences. This does not change the estimand; it only provides the conventional regression-style coefficient, standard error, and p-value before the design-based RI calculation.

Table 33: Bridge linear model for the paired estimand

quantity	value
Intercept (mean paired effect)	14.3694
Standard error	4.6916
t statistic	3.0628
Conventional two-sided p-value	0.0041
95% CI lower	4.8633
95% CI upper	23.8756

0.47 Randomization distribution

Table 34: Randomization inference summary

quantity	value
Simulations	10000.0000
Observed paired mean difference	14.3694
Two-sided RI p-value	0.0031
One-sided RI p-value (high > low)	0.0014

0.48 Figure

Randomization distribution under the sharp null

Solid orange = observed statistic; dashed orange = symmetric reference

